

Week - 05 - Classes, Graphics\Lecture_10_code_3.py

```
1  '''
2  This code demonstrates associations between Canon and Projectile classes via composition
   mechanism.
3  Class Canon creates its own Projectile object in its constructor self.bomb =
   Projectile(201, 25, 12). This Projectile object cannot lives without Canon object. .
4  '''
5  import math
6  import time
7
8  G = 9.82
9  AIR_DRAG = 0.4
10
11 class Projectile:
12     def __init__(self, ID, size, energy, x_pos=0, y_pos=0):
13         self.id = ID
14         self.size = size
15         self.energy = energy
16         self.x_pos = x_pos
17         self.y_pos = y_pos
18
19     def get_projectile_data(self):
20         # A getter function
21         return (f'id {self.id}, size {self.size}, energy {self.energy} and position
   ({self.x_pos}, {self.y_pos})')
22
23
24 class Canon:
25     # Now a composition design
26     def __init__(self, cid, length, type):
27         self.cid = cid
28         self.length = length
29         self.type = type
30         self.bomb = Projectile(201, 25, 12)
31
32 # def method_name (self, another_object)
33     def fire(self, velocity, theta):
34         x_vel = velocity * math.cos(theta)
35         y_vel = velocity * math.sin(theta)
36         dt = 0.02
37
38         while self.bomb.y_pos >= 0:
39             self.bomb.x_pos += x_vel * dt
40             self.bomb.y_pos += y_vel * dt
41             x_vel -= AIR_DRAG * dt
42             y_vel -= G * dt
43             time.sleep(0.2)
44
45             animate(self.bomb.x_pos, self.bomb.y_pos)
46
47     def load_projectile(self):
48         self.bomb.x_pos = 0
49
```

```
50 # Helper function to display trajectory of our projectile
51 def animate(x, y):
52     width, height = 60, 30
53     scale = 100
54     screen = [[" " for _ in range(width)] for _ in range(height)]
55
56     column = round(x * scale)
57     row = height - 1 - round(y * scale)
58     if 0 <= column < width and 0 <= row < height:
59         screen[row][column] = "*"
60
61     print("\033[2J\033[H", end="")
62     print("\n".join("".join(line) for line in screen), flush=True)
63     time.sleep(0.1)
64
65
66 gun_1 = Canon('Gun-13', 20, 'Ground-based muzzle')
67 gun_1.fire(2, math.radians(45))
```